

# **BANGLADESH RESEARCH AND EDUCATION NETWORK**

## **NOC MONTHLY REPORT**

APRIL | 2021

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1. COORDINATION ISSUES

Processed 661 mail updates between BdREN NOC and member institution.

2. LINK AVAILABILITY

| Name of Institutes | Outage (Minute) | Maintenance (Minute) | Availability (w/o Maintenance) | Availability (with Maintenance) |
|--------------------|-----------------|----------------------|--------------------------------|---------------------------------|
| AUB                |                 |                      | 100.00%                        | 100.00%                         |
| AUST               |                 | 195                  | 100.00%                        | 99.56%                          |
| BARI               |                 |                      | 100.00%                        | 100.00%                         |
| BAU                |                 |                      | 100.00%                        | 100.00%                         |
| BFRI               |                 |                      | 100.00%                        | 100.00%                         |
| BLRI               |                 |                      | 100.00%                        | 100.00%                         |
| BoU                |                 |                      | 100.00%                        | 100.00%                         |
| BRAC               |                 |                      | 100.00%                        | 100.00%                         |
| BRRI               |                 |                      | 100.00%                        | 100.00%                         |
| BRUR               | 194             |                      | 99.57%                         | 99.57%                          |
| BSMMU              |                 |                      | 100.00%                        | 100.00%                         |
| BSMRAU             |                 |                      | 100.00%                        | 100.00%                         |
| BSMRSTU            |                 |                      | 100.00%                        | 100.00%                         |
| BU                 |                 |                      | 100.00%                        | 100.00%                         |
| BUET               |                 |                      | 100.00%                        | 100.00%                         |
| BUP                |                 |                      | 100.00%                        | 100.00%                         |
| BUTex              |                 |                      | 100.00%                        | 100.00%                         |
| CoU                |                 |                      | 100.00%                        | 100.00%                         |
| CU                 |                 | 195                  | 100.00%                        | 99.56%                          |
| CUET               |                 | 195                  | 100.00%                        | 99.56%                          |
| CVASU              |                 | 195                  | 100.00%                        | 99.56%                          |
| DIU                |                 | 235                  | 100.00%                        | 99.47%                          |
| DU                 |                 |                      | 100.00%                        | 100.00%                         |
| DUET               |                 |                      | 100.00%                        | 100.00%                         |
| EU                 |                 |                      | 100.00%                        | 100.00%                         |
| HSTU               | 194             |                      | 99.57%                         | 99.57%                          |
| ICMAB              |                 |                      | 100.00%                        | 100.00%                         |
| IU                 |                 |                      | 100.00%                        | 100.00%                         |
| IUB                |                 |                      | 100.00%                        | 100.00%                         |
| IUT                |                 |                      | 100.00%                        | 100.00%                         |
| JKKNIU             |                 |                      | 100.00%                        | 100.00%                         |
| JnU                |                 |                      | 100.00%                        | 100.00%                         |
| JU                 |                 |                      | 100.00%                        | 100.00%                         |
| JUST               |                 |                      | 100.00%                        | 100.00%                         |
| KU                 |                 |                      | 100.00%                        | 100.00%                         |
| KUET               |                 |                      | 100.00%                        | 100.00%                         |
| LUS                |                 |                      | 100.00%                        | 100.00%                         |
| MBSTU              | 255             |                      | 99.43%                         | 99.43%                          |
| MIST               |                 |                      | 100.00%                        | 100.00%                         |
| NINMAS             | 1171            |                      | 96.15%                         | 96.15%                          |

| Name of Institutes | Outage (Minute) | Maintenance (Minute) | Availability (w/o Maintenance) | Availability (with Maintenance) |
|--------------------|-----------------|----------------------|--------------------------------|---------------------------------|
| NSTU               | 685             |                      | 98.47%                         | 98.03%                          |
| NSU                |                 |                      | 100.00%                        | 100.00%                         |
| NU                 |                 |                      | 100.00%                        | 100.00%                         |
| PSTU               |                 |                      | 100.00%                        | 100.00%                         |
| PUST               |                 |                      | 100.00%                        | 100.00%                         |
| RU                 |                 |                      | 100.00%                        | 100.00%                         |
| RUET               |                 |                      | 100.00%                        | 100.00%                         |
| SAU                | 92              |                      | 99.79%                         | 99.79%                          |
| SBAU               |                 |                      | 100.00%                        | 100.00%                         |
| SUST               |                 |                      | 100.00%                        | 100.00%                         |

Table 1: Institution Wise Link Availability

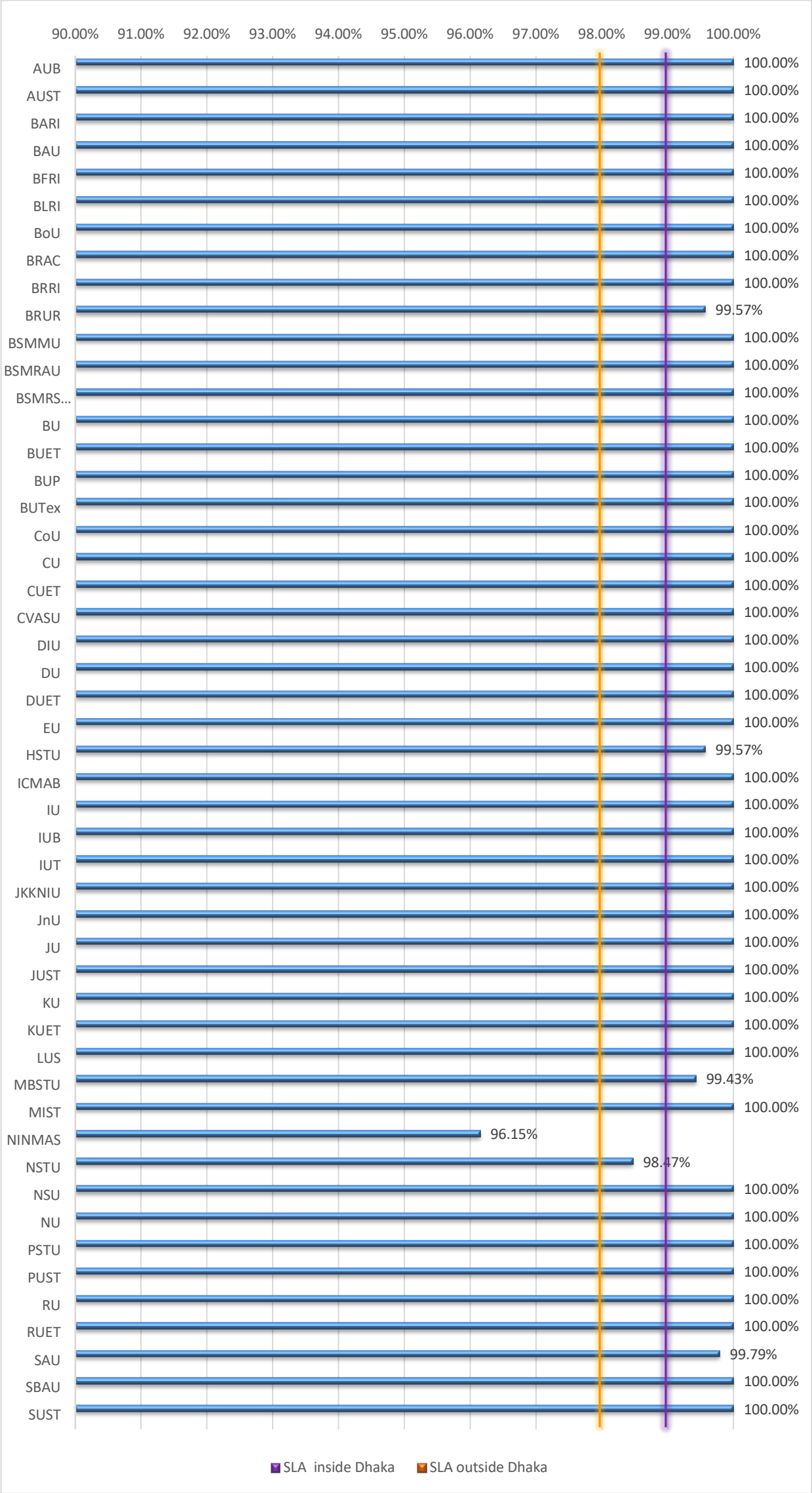


Figure 1: Institution Wise Link Availability w/o Maintenance

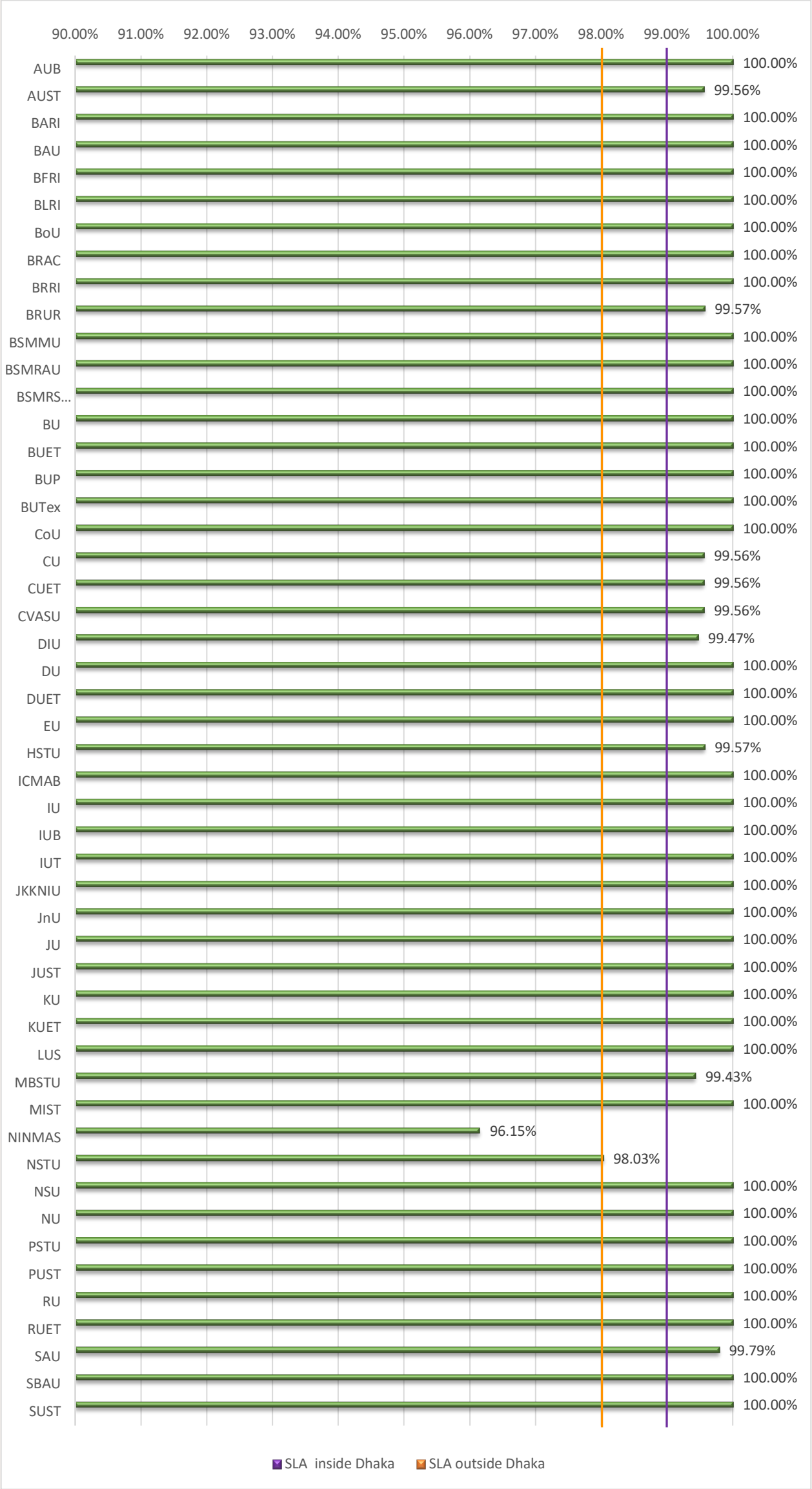


Figure 2: Institution Wise Link Availability with Maintenance

### 3. MAJOR OUTAGE OF APRIL, 2021

There were 4 link/Node outages.

- ❖ Ticket#2021040601000011
  - Affected Member** : BRUR, HSTU
  - Recovery Time** : 194 min
  - SLA Compliance** : Fulfilled
  - Problem Details** : There was an underground fiber break about 800m away from the university's main gate. The BdREN team restored the fiber and the link went up.
- ❖ Ticket#2021041701000027
  - Affected Member** : NINMAS
  - Recovery Time** : 1171 min
  - SLA Compliance** : Not Fulfilled [Force Majeure]
  - Problem Details** : Since we couldn't get access permission to NINMAS, the link troubleshooting was delayed. After getting access, the router was reconfigured, and the service was restored.
- ❖ Ticket#2021042301000024
  - Affected Member** : NSTU
  - Recovery Time** : 685 min
  - SLA Compliance** : Fulfilled
  - Problem Details** : A fiber break occurred about 10.40 kilometers away from NSTU. Our transmission team was able to successfully repair the damaged underground fiber.
- ❖ Ticket#2021041901000014
  - Affected Member** : SAU
  - Recovery Time** : 132 min
  - SLA Compliance** : Fulfilled
  - Problem Details** : The SAU Node was down due to a power related issue. After initial troubleshooting, the BdREN team found a problem with the Magnetic Contactor (MC) of ATS. The BdREN team disconnected the commercial power, and the site was temporarily powered by the generator. Later, the maintenance team resolved the matter replacing the broken equipment.

### 4. TRAFFIC STATISTICS

| INSTITUTION | TOTAL TRAFFIC |         | AVERAGE BW |            | MAXIMUM BW |            |
|-------------|---------------|---------|------------|------------|------------|------------|
|             | IN            | OUT     | IN         | OUT        | IN         | OUT        |
| DU          | 462.3 TB      | 99.2 TB | 542.0 Mbps | 143.0 Mbps | 949.6 Mbps | 619.6 Mbps |
| BUET        | 229.5 TB      | 42.3 TB | 301.5 Mbps | 44.6 Mbps  | 667.5 Mbps | 116.4 Mbps |
| BAU         | 204.9 TB      | 30.7 TB | 254.0 Mbps | 63.0 Mbps  | 429.5 Mbps | 113.2 Mbps |
| RU          | 120.3 TB      | 24.3 TB | 151.0 Mbps | 26.0 Mbps  | 308.0 Mbps | 174.0 Mbps |
| JU          | 119.2 TB      | 11.5 TB | 161.0 Mbps | 21.0 Mbps  | 345.7 Mbps | 71.2 Mbps  |
| CU          | 85.5 TB       | 16.9 TB | 95.0 Mbps  | 10.0 Mbps  | 230.0 Mbps | 52.0 Mbps  |
| SBAU        | 81.5 TB       | 6.5 TB  | 95.0 Mbps  | 10.0 Mbps  | 216.0 Mbps | 57.5 Mbps  |
| CUET        | 69.8 TB       | 6.2 TB  | 85.2 Mbps  | 8.8 Mbps   | 230.5 Mbps | 66.7 Mbps  |
| BSMMU       | 50.4 TB       | 4.9 TB  | 71.0 Mbps  | 6.9 Mbps   | 240.4 Mbps | 40.6 Mbps  |
| DUET        | 44.9 TB       | 5.8 TB  | 51.0 Mbps  | 4.6 Mbps   | 163.1 Mbps | 29.9 Mbps  |
| JUST        | 44.0 TB       | 5.8 TB  | 64.0 Mbps  | 11.0 Mbps  | 297.6 Mbps | 95.6 Mbps  |
| CoU         | 42.0 TB       | 2.6 TB  | 88.7 Mbps  | 6.5 Mbps   | 209.3 Mbps | 32.4 Mbps  |
| HSTU        | 30.6 TB       | 1.9 TB  | 29.0 Mbps  | 3.0 Mbps   | 76.5 Mbps  | 13.6 Mbps  |
| RUET        | 26.2 TB       | 1.9 TB  | 44.0 Mbps  | 3.2 Mbps   | 219.2 Mbps | 27.2 Mbps  |
| KUET        | 25.8 TB       | 3.7 TB  | 23.0 Mbps  | 3.9 Mbps   | 148.1 Mbps | 38.1 Mbps  |

|         |         |         |            |            |            |            |
|---------|---------|---------|------------|------------|------------|------------|
| IUT     | 24.4 TB | 13.3 TB | 131.0 Mbps | 27.0 Mbps  | 189.0 Mbps | 69.8 Mbps  |
| JKKNIU  | 24.2 TB | 1.7 TB  | 24.0 Mbps  | 2.1 Mbps   | 85.4 Mbps  | 10.2 Mbps  |
| IU      | 23.8 TB | 1.3 TB  | 35.7 Mbps  | 1.9 Mbps   | 113.9 Mbps | 30.5 Mbps  |
| BRUR    | 22.9 TB | 1.7 TB  | 24.0 Mbps  | 4.0 Mbps   | 136.1 Mbps | 35.7 Mbps  |
| CVASU   | 20.9 TB | 1.5 TB  | 23.3 Mbps  | 1.5 Mbps   | 88.9 Mbps  | 10.6 Mbps  |
| BOU     | 20.7 TB | 19.1 TB | 21.0 Mbps  | 9.0 Mbps   | 90.7 Mbps  | 44.1 Mbps  |
| PSTU    | 19.3 TB | 0.9 TB  | 38.0 Mbps  | 3.2 Mbps   | 132.3 Mbps | 16.8 Mbps  |
| MBSTU   | 16.0 TB | 1.9 TB  | 33.0 Mbps  | 3.7 Mbps   | 175.4 Mbps | 25.4 Mbps  |
| JnU     | 15.4 TB | 1.1 TB  | 15.0 Mbps  | 1.5 Mbps   | 105.1 Mbps | 23.5 Mbps  |
| SUST    | 14.4 TB | 6.5 TB  | 43.0 Mbps  | 33.0 Mbps  | 186.6 Mbps | 59.6 Mbps  |
| BUTex   | 14.1 TB | 1.5 TB  | 13.0 Mbps  | 1.4 Mbps   | 159.7 Mbps | 41.5 Mbps  |
| NSTU    | 13.3 TB | 3.3 TB  | 25.1 Mbps  | 7.0 Mbps   | 120.6 Mbps | 22.6 Mbps  |
| IUB     | 13.3 TB | 0.8 TB  | 12.0 Mbps  | 0.6 Mbps   | 118.4 Mbps | 35.8 Mbps  |
| BU      | 9.4 TB  | 0.5 TB  | 11.6 Mbps  | 0.7 Mbps   | 65.6 Mbps  | 6.0 Mbps   |
| NU      | 8.2 TB  | 1.7 TB  | 13.1 Mbps  | 7.4 Mbps   | 60.5 Mbps  | 38.9 Mbps  |
| BSMRAU  | 6.7 TB  | 1.6 TB  | 10.0 Mbps  | 7.2 Mbps   | 76.9 Mbps  | 23.4 Mbps  |
| BLRI    | 5.8 TB  | 0.6 TB  | 24.1 Mbps  | 2.3 Mbps   | 62.1 Mbps  | 8.8 Mbps   |
| SAU     | 5.8 TB  | 0.4 TB  | 25.0 Mbps  | 1.9 Mbps   | 67.9 Mbps  | 5.7 Mbps   |
| KU      | 5.7 TB  | 0.7 TB  | 30.7 Mbps  | 4.0 Mbps   | 102.7 Mbps | 10.3 Mbps  |
| DIU     | 5.7 TB  | 49.3 TB | 41.7 Mbps  | 265.0 Mbps | 119.6 Mbps | 508.2 Mbps |
| BRRI    | 5.1 TB  | 0.4 TB  | 16.9 Mbps  | 3.0 Mbps   | 47.5 Mbps  | 11.3 Mbps  |
| PUST    | 4.0 TB  | 0.3 TB  | 24.0 Mbps  | 2.0 Mbps   | 105.2 Mbps | 12.1 Mbps  |
| BSMRSTU | 4.0 TB  | 0.2 TB  | 23.9 Mbps  | 3.0 Mbps   | 69.9 Mbps  | 9.1 Mbps   |
| BARI    | 3.5 TB  | 0.3 TB  | 12.5 Mbps  | 1.3 Mbps   | 82.8 Mbps  | 15.7 Mbps  |
| BFRI    | 1.3 TB  | 0.1 TB  | 1.5 Mbps   | 0.2 Mbps   | 10.8 Mbps  | 9.1 Mbps   |
| EU      | 1.1 TB  | 3.4 TB  | 12.5 Mbps  | 11.0 Mbps  | 35.9 Mbps  | 26.5 Mbps  |
| EWU     | 0.4 TB  | 0.0 TB  | 3.5 Mbps   | 0.1 Mbps   | 31.4 Mbps  | 3.1 Mbps   |
| AUST    | 0.3 TB  | 0.9 TB  | 6.3 Mbps   | 0.9 Mbps   | 44.5 Mbps  | 9.3 Mbps   |
| MIST    | 0.1 TB  | 0.0 TB  | 8.6 Mbps   | 3.0 Mbps   | 26.4 Mbps  | 32.3 Mbps  |
| BUP     | 0.1 TB  | 0.0 TB  | 0.1 Mbps   | 0.0 Mbps   | 1.6 Mbps   | 0.7 Mbps   |
| NSU     | 0.0 TB  | 0.0 TB  | 0.2 Mbps   | 0.1 Mbps   | 3.1 Mbps   | 0.7 Mbps   |
| BRACU   | 0.0 TB  | 0.0 TB  | 2.6 Mbps   | 0.9 Mbps   | 20.0 Mbps  | 3.6 Mbps   |

Table 2: Institution Wise Traffic Statistics

5. TOTAL TRAFFIC

|                |            |
|----------------|------------|
| Total Download | 1946.53 TB |
| Total Upload   | 379.45 TB  |

6. ONLINE EDUCATION

6.1 Institution Wise vSession Usage Statistics

➤ Public Universities and Colleges:

| SL | University Name               | Total_Class_Taken | Total_Hours | Total_Participants |
|----|-------------------------------|-------------------|-------------|--------------------|
| 1  | Jahangirnagar University (JU) | 2456              | 3114        | 117964             |
| 2  | Lalmatia Mohila College       | 1989              | 1398        | 29763              |
| 3  | University of Rajshahi (RU)   | 1897              | 2119        | 88788              |
| 4  | University of Chittagong (CU) | 1829              | 2535        | 123969             |
| 5  | Jagannath University (JnU)    | 1706              | 2141        | 62013              |
| 6  | University of Dhaka (DU)      | 1641              | 2166        | 69202              |
| 7  | Cumilla University (CoU)      | 1125              | 1342        | 54064              |



| SL | University Name                                                                | Total_Class_Taken | Total_Hours | Total_Participants |
|----|--------------------------------------------------------------------------------|-------------------|-------------|--------------------|
| 8  | Bangabandhu Sheikh Mujibur Rahman Agricultural University (BSMRAU)             | 898               | 825         | 52070              |
| 9  | Bangladesh Agricultural University (BAU)                                       | 887               | 1046        | 61506              |
| 10 | Sher-e-Bangla Agricultural University (SBAU)                                   | 839               | 779         | 80505              |
| 11 | Bangladesh University of Textiles (BUTEX)                                      | 767               | 1025        | 36895              |
| 12 | Islamic University of Technology (IUT)                                         | 730               | 820         | 29417              |
| 13 | Hajee Mohammad Danesh Science & Technology University (HSTU)                   | 719               | 837         | 59124              |
| 14 | Noakhali Science and Technology University (NSTU)                              | 716               | 872         | 30244              |
| 15 | University of Barishal (BU)                                                    | 600               | 655         | 30635              |
| 16 | Military Institute of Science and Technology (MIST)                            | 575               | 753         | 32797              |
| 17 | Cumilla Shikkha Board Govt. Model College                                      | 457               | 273         | 12591              |
| 18 | Mawlana Bhashani Science and Technology University (MBSTU)                     | 452               | 488         | 15467              |
| 19 | Begum Rokeya University (BRUR)                                                 | 426               | 483         | 29076              |
| 20 | National Institute of Textile Engineering and Research (NITER)                 | 330               | 284         | 14311              |
| 21 | Bangabandhu Sheikh Mujib Medical University (BSMMU)                            | 325               | 485         | 11160              |
| 22 | Dhaka University of Engineering & Technology (DUET)                            | 315               | 398         | 40742              |
| 23 | Bangladesh Open University (BOU)                                               | 310               | 409         | 10228              |
| 24 | Islamic University, Bangladesh (IU)                                            | 300               | 356         | 19436              |
| 25 | Patuakhali Science and Technology University (PSTU)                            | 297               | 309         | 16217              |
| 26 | Jashore University of Science and Technology (JUST)                            | 288               | 334         | 8663               |
| 27 | Khulna University (KU)                                                         | 278               | 372         | 7934               |
| 28 | Bangladesh University of Professionals (BUP)                                   | 271               | 423         | 18489              |
| 29 | Chittagong University of Engineering & Technology (CUET)                       | 270               | 326         | 22020              |
| 30 | Pabna University of Science and Technology (PUST)                              | 264               | 321         | 12842              |
| 31 | Bangabandhu Sheikh Mujibur Rahman Science and Technology University (BSMRSTU)  | 260               | 282         | 17475              |
| 32 | Khulna University of Engineering & Technology (KUET)                           | 253               | 288         | 16660              |
| 33 | Jatiya Kabi Kazi Nazrul Islam University (JKKNIU)                              | 248               | 317         | 9916               |
| 34 | Bangladesh University of Engineering & Technology (BUET)                       | 248               | 321         | 11169              |
| 35 | Bangabandhu Sheikh Mujibur Rahman Maritime University                          | 236               | 324         | 4729               |
| 36 | Rangamati Science and Technology University (RMSTU)                            | 149               | 186         | 5870               |
| 37 | Cumilla Victoria Government College                                            | 133               | 136         | 2761               |
| 38 | Chittagong Veterinary and Animal Sciences University (CVASU)                   | 123               | 127         | 6981               |
| 39 | Khulna Agricultural University (KAU)                                           | 111               | 138         | 3794               |
| 40 | Bangamata Sheikh Fojilatunnesa Mujib Science & Technology University (BSFMSTU) | 92                | 102         | 4549               |
| 41 | Shahjalal University of Science and Technology (SUST)                          | 92                | 120         | 2055               |
| 42 | Sheikh Hasina University (SHU)                                                 | 83                | 100         | 2649               |

| SL | University Name                                                               | Total_Class_Taken | Total_Hours | Total_Participants |
|----|-------------------------------------------------------------------------------|-------------------|-------------|--------------------|
| 43 | Mymensingh Engineering College                                                | 75                | 77          | 2175               |
| 44 | Sylhet Agricultural University (SAU)                                          | 63                | 84          | 2073               |
| 45 | Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University (BSMRAAU) | 61                | 97          | 2645               |
| 46 | National University, Bangladesh                                               | 59                | 92          | 1325               |
| 47 | Eden Mohila College                                                           | 53                | 48          | 1600               |
| 48 | Chandpur Government College, Chandpur                                         | 51                | 26          | 412                |
| 49 | Mohammadpur Govt college, Dhaka                                               | 43                | 54          | 521                |
| 50 | Rabindra University, Bangladesh (RUB)                                         | 36                | 46          | 1956               |
| 51 | Rajshahi University of Engineering & Technology (RUET)                        | 34                | 32          | 2254               |
| 52 | Faridpur Engineering College                                                  | 19                | 16          | 350                |
| 53 | Bangabandhu Digital University                                                | 1                 | 1           | 13                 |

Table 3: vSession Usage by Public Universities and Colleges

➤ Private Universities:

| SL | Institution Name                                                        | Total Class | Total Hours | Total Participants |
|----|-------------------------------------------------------------------------|-------------|-------------|--------------------|
| 1  | International Islamic University Chittagong (IIUC)                      | 6811        | 7907.1868   | 248113             |
| 2  | Green University of Bangladesh                                          | 5909        | 7764.2845   | 245115             |
| 3  | Primeasia University                                                    | 2876        | 3429.0675   | 81290              |
| 4  | Ahsanullah University of Science and Tech.                              | 2837        | 3204.4527   | 72252              |
| 5  | Port City International University                                      | 2406        | 3153.6846   | 86452              |
| 6  | Shanto-Mariam University of Creative Technology                         | 2366        | 3587.4313   | 71898              |
| 7  | Stamford University Bangladesh                                          | 2363        | 3210.2329   | 68387              |
| 8  | Bangladesh Army University of Engineering and Technology                | 2097        | 1790.9332   | 53617              |
| 9  | Leading University, Sylhet                                              | 1955        | 2853.2005   | 53639              |
| 10 | State University of Bangladesh                                          | 1889        | 2979.667    | 56975              |
| 11 | Premier University, Chittagong                                          | 1820        | 2078.8835   | 70606              |
| 12 | Dhaka International University (DIU)                                    | 1770        | 1766.7155   | 58311              |
| 13 | BGMEA University of Fashion & Technology                                | 1640        | 2041.8493   | 77166              |
| 14 | Eastern University                                                      | 1567        | 1737.6999   | 58807              |
| 15 | Asian University of Bangladesh                                          | 1560        | 1936.568    | 40019              |
| 16 | Bangladesh Army International University of Science and Technology      | 1446        | 1525.7008   | 43892              |
| 17 | BGC Trust University Bangladesh                                         | 1337        | 1295.7832   | 32860              |
| 18 | ASA University Bangladesh                                               | 1228        | 1785.1828   | 20130              |
| 19 | North Western University (Khulna)                                       | 1185        | 1018.4505   | 25627              |
| 20 | International University of Business Agriculture and Technology (IUBAT) | 981         | 1132.7167   | 44641              |
| 21 | Prime University                                                        | 895         | 1013.6668   | 24263              |
| 22 | City University                                                         | 875         | 1240.7331   | 36746              |
| 23 | University of Science and Technology Chittagong (USTC)                  | 864         | 884.5004    | 24747              |
| 24 | Manarat International University                                        | 829         | 1034.8835   | 28177              |
| 25 | Gono Bishwabidyalay                                                     | 669         | 618.9168    | 25976              |
| 26 | University of Asia Pacific                                              | 509         | 877.3675    | 20315              |
| 27 | Bangladesh Islami University                                            | 508         | 544.6666    | 8505               |
| 28 | Feni University                                                         | 491         | 613.9827    | 10906              |
| 29 | University of Information Technology and Sciences                       | 459         | 541.2496    | 13777              |
| 30 | Presidency University                                                   | 420         | 484.3332    | 11107              |
| 31 | Bangladesh University of Health Sciences                                | 413         | 504.7833    | 11011              |
| 32 | European University of Bangladesh                                       | 383         | 357.3673    | 13043              |

| SL | Institution Name                                | Total Class | Total Hours | Total Participants |
|----|-------------------------------------------------|-------------|-------------|--------------------|
| 33 | Northern Univ. of Business & Tech Khulna        | 364         | 474.7998    | 12203              |
| 34 | BRAC University                                 | 332         | 428.6167    | 9387               |
| 35 | CCN University of Science & Technology, Cumilla | 321         | 284.6342    | 6131               |
| 36 | Notre Dame University Bangladesh                | 319         | 369.9659    | 7891               |
| 37 | Southeast University                            | 289         | 409.7172    | 11864              |
| 38 | University of South Asia                        | 250         | 275.567     | 9238               |
| 39 | North East University Bangladesh                | 237         | 273.7002    | 6410               |
| 40 | Atish Dipankar University of Science and Tech   | 195         | 205.7334    | 4202               |
| 41 | Pundra University of Science & Technology       | 189         | 188.2675    | 5135               |
| 42 | Adamjee Cantonment College                      | 170         | 174.4999    | 2883               |
| 43 | University of Liberal Arts Bangladesh (ULAB)    | 130         | 208.5665    | 4840               |
| 44 | United International University                 | 98          | 147.3668    | 3845               |
| 45 | East West University (EWU)                      | 59          | 126.1496    | 1889               |
| 46 | International Standard University               | 59          | 66.3328     | 1751               |
| 47 | East Delta University                           | 25          | 23.6832     | 832                |
| 48 | University of Liberal Arts Bangladesh           | 23          | 39.3166     | 1580               |
| 49 | North South University                          | 23          | 43.1335     | 844                |
| 50 | Khwaja Yunus Ali University                     | 21          | 40.3834     | 1602               |
| 51 | Daffodil International University               | 17          | 27.0498     | 1762               |
| 52 | Bangladesh University                           | 17          | 27.0336     | 101                |
| 53 | Metro University                                | 10          | 15.2167     | 219                |
| 54 | Notre Dame College, Dhaka                       | 9           | 11.7168     | 459                |
| 55 | Varendra University                             | 5           | 7.2666      | 310                |
| 56 | World University                                | 3           | 4.35        | 198                |
| 57 | Ranada Prasad Shaha University (RPSU)           | 2           | 2.2334      | 245                |

Table 4: vSession Usage by Private University

➤ Medical Colleges:

| SL | Institution Name                         | Total Class | Total Hours | Total Participants |
|----|------------------------------------------|-------------|-------------|--------------------|
| 1  | TMSS Medical College                     | 536         | 590         | 58045              |
| 2  | Ibrahim Medical College, Dhaka           | 407         | 582         | 27169              |
| 3  | Dhaka Medical College                    | 231         | 300         | 24048              |
| 4  | Holy Family Red Crescent Medical College | 226         | 203         | 9664               |
| 5  | North East Medical College               | 145         | 144         | 14475              |
| 6  | Faridpur Medical College                 | 145         | 153         | 13899              |
| 7  | MAG Osmani Medical College, Sylhet       | 19          | 24          | 1531               |
| 8  | Anwar Khan Modern Medical College        | 8           | 15          | 356                |
| 9  | Kumudini Women's Medical College         | 8           | 2           | 13                 |
| 10 | Rajshahi Medical University              | 3           | 3           | 66                 |

Figure 3: vSession Usage by Medical Colleges

➤ Other Institution:

| SL | Institution Name                                        | Total Class | Total Hours | Total Participants |
|----|---------------------------------------------------------|-------------|-------------|--------------------|
| 1  | Bangladesh Institute of Management (BIM)                | 242         | 373.8164    | 14761              |
| 2  | Institute of Chartered Secretaries of Bangladesh (ICSB) | 100         | 222.8827    | 5590               |
| 3  | Bangladesh Institute of Capital Market                  | 46          | 54.9333     | 484                |
| 4  | Bangladesh Association of Women Scientist (BAWS)        | 15          | 27.8666     | 748                |
| 5  | Bangladesh Livestock Research Institute-BLRI            | 7           | 8.7333      | 635                |
| 6  | Bangladesh Research and Education Network (BdREN)       | 4           | 9.3         | 40                 |

Table 5: vSession Usage by Other Institution

6.2 Institution Wise vSession Usage Graph

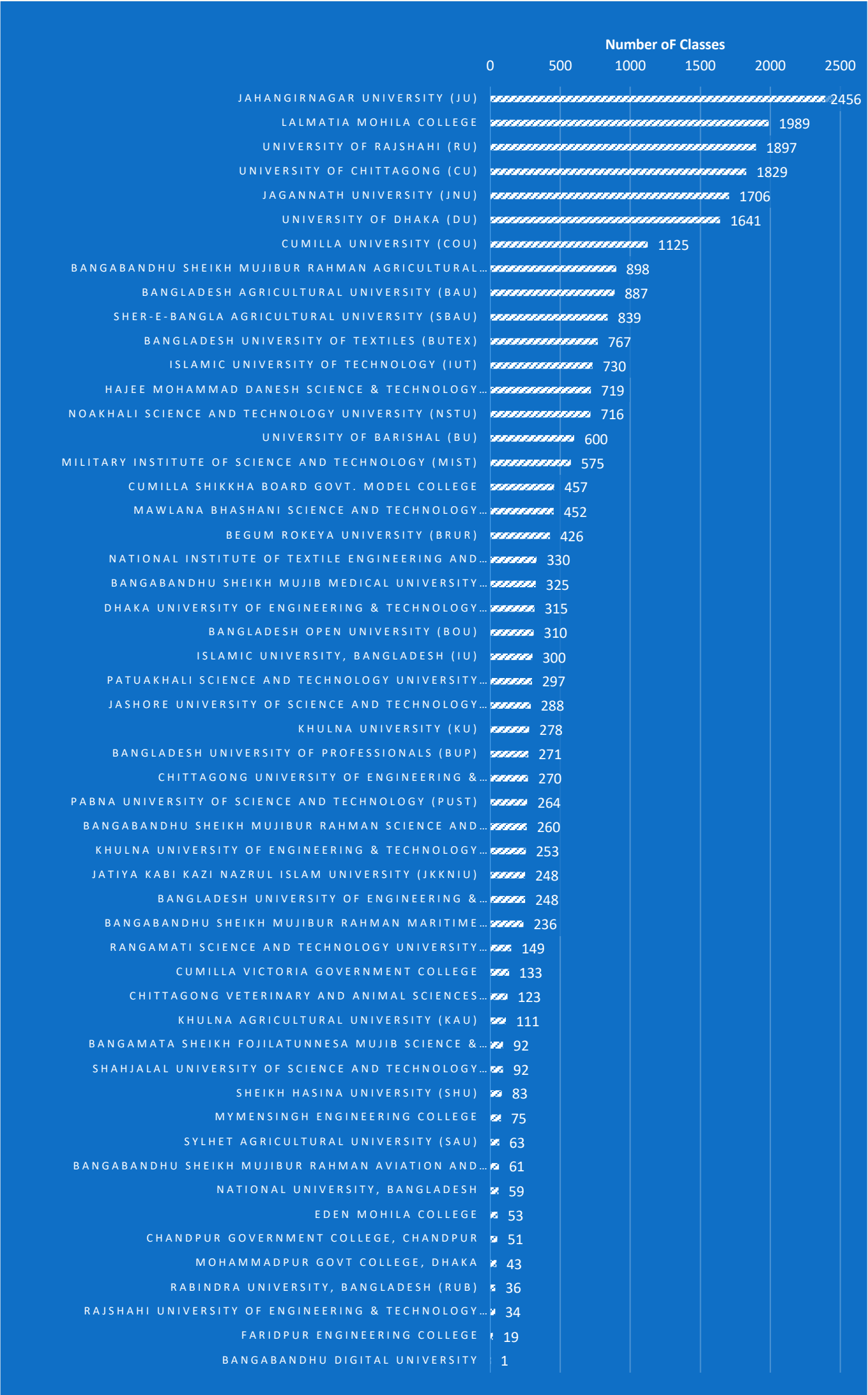


Figure 4:Public University & Colleges vSession Usage Graph

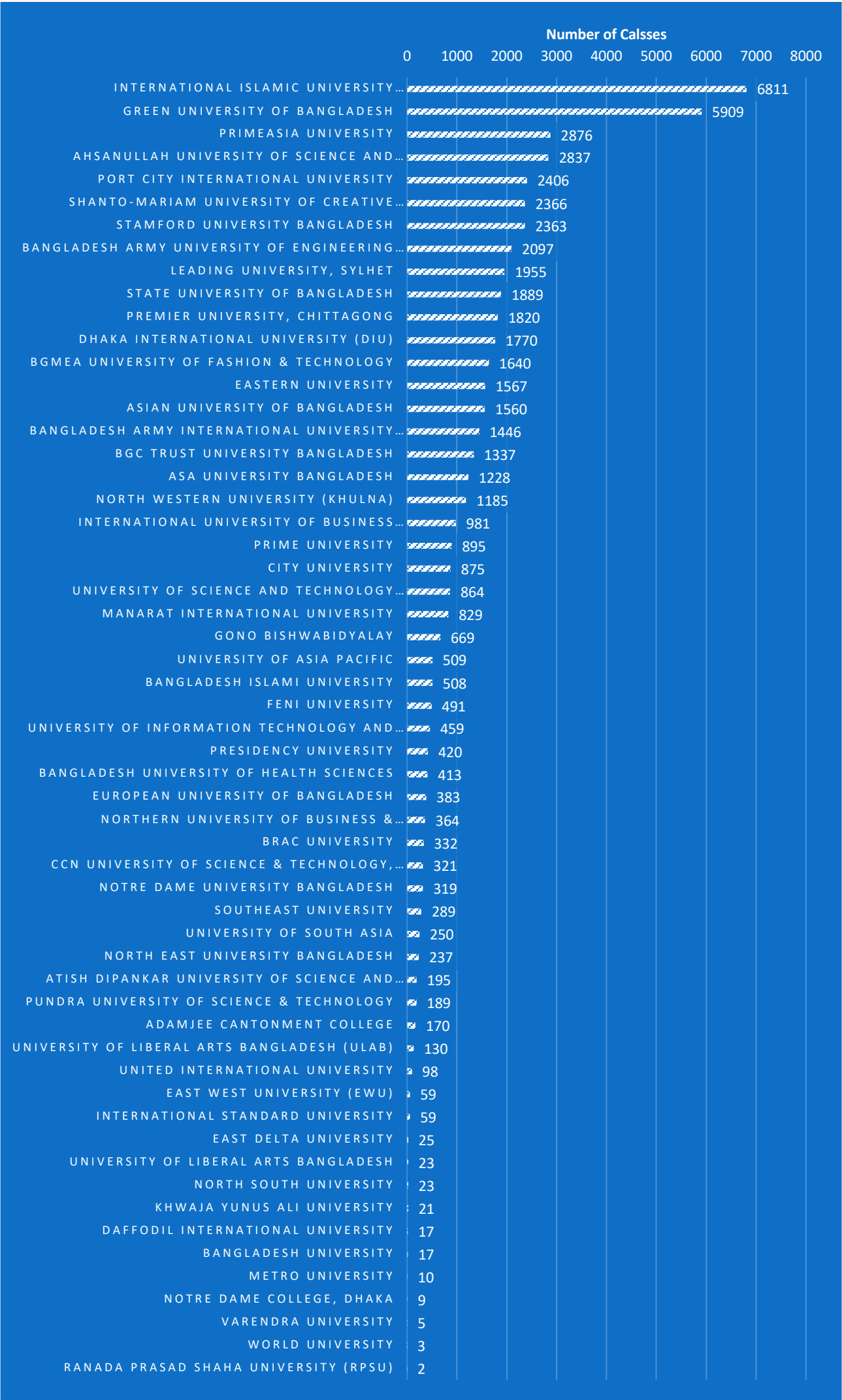


Figure 5:Private University vSession Usage Graph

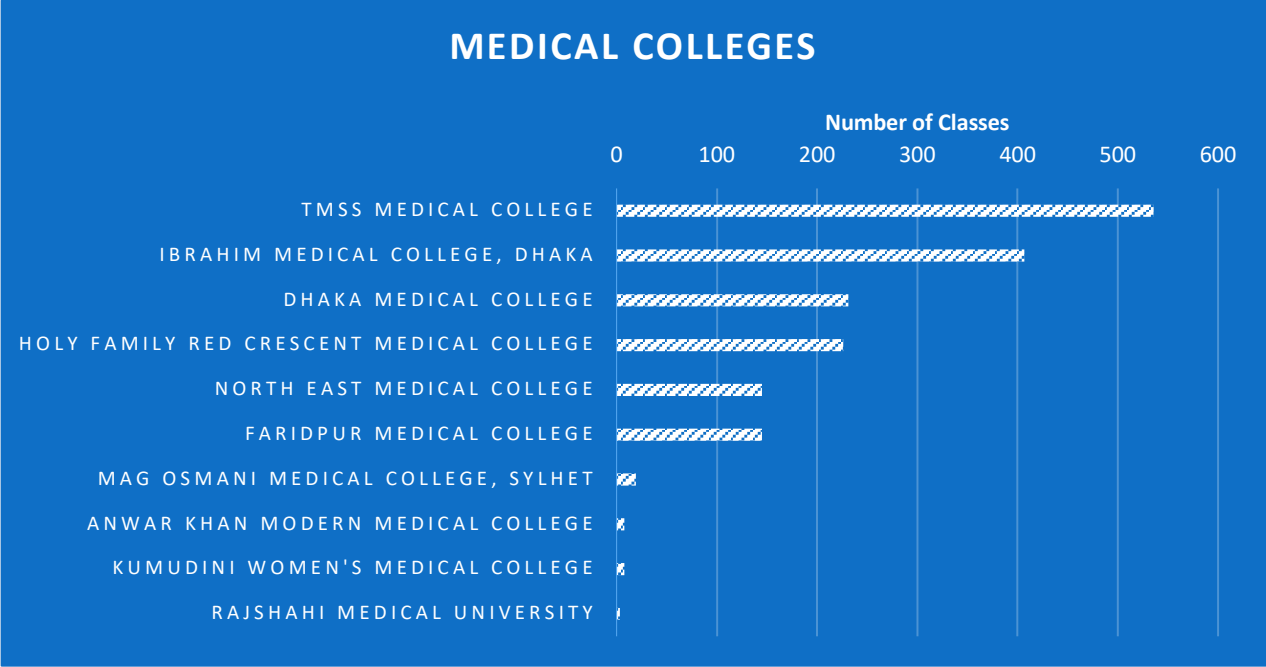


Figure 6: Medical College vSession Usage Graph

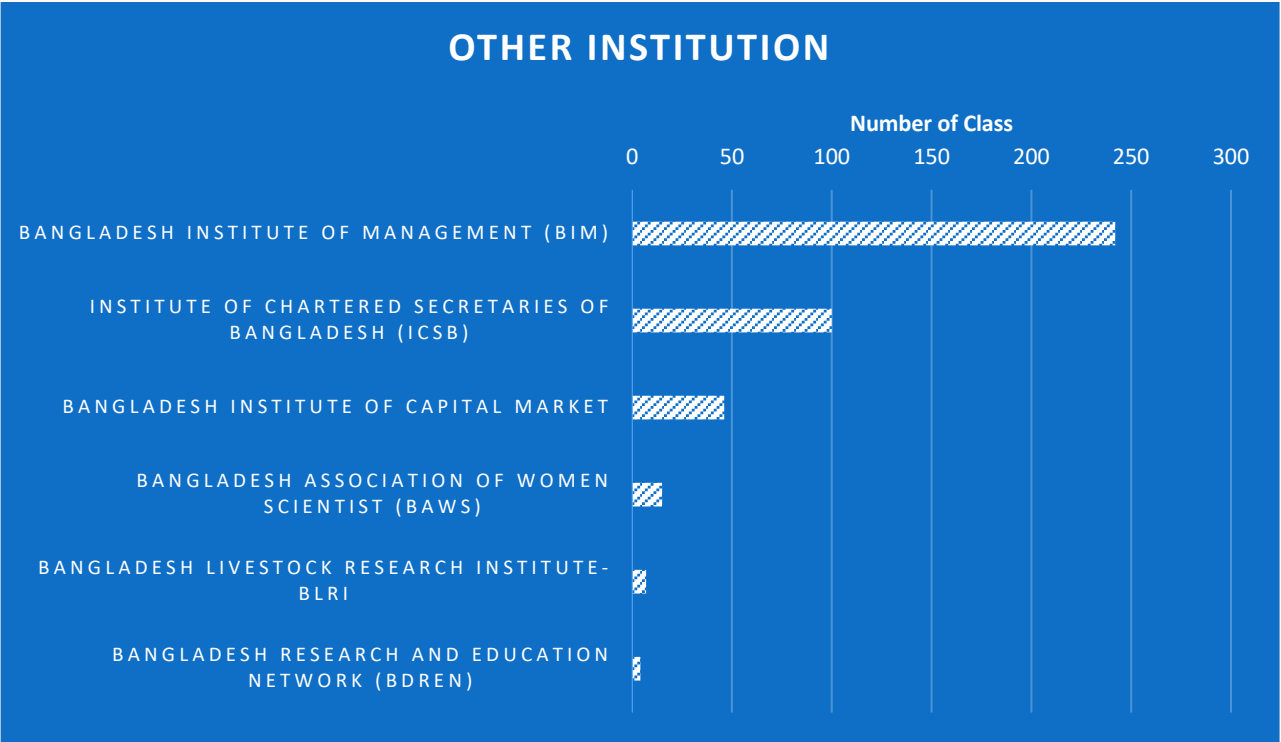


Figure 7: Other Institution vSession Usage Graph

| 6.3 Institution Wise On-Prem Zoom Usage Statistics |                                                   |                   |                   |                    |
|----------------------------------------------------|---------------------------------------------------|-------------------|-------------------|--------------------|
| SI                                                 | Institution Name                                  | Number Of Classes | Total Class Hours | Total Participants |
| 1                                                  | University of Dhaka                               | 6397              | 8579:43:53        | 314473             |
| 2                                                  | BUET                                              | 3462              | 4363:16:37        | 108946             |
| 3                                                  | Military Institute of Science and Technology      | 3349              | 4718:39:23        | 233588             |
| 4                                                  | Bangladesh University of Professionals            | 2221              | 3012:51:06        | 190702             |
| 5                                                  | Notre Dame College, Dhaka                         | 2179              | 1725:55:08        | 510502             |
| 6                                                  | KUET                                              | 2050              | 2410:06:34        | 142752             |
| 7                                                  | CUET                                              | 1511              | 1743:09:19        | 110805             |
| 8                                                  | United International University (UIU)             | 1480              | 2061:34:50        | 69947              |
| 9                                                  | Dhaka University of Engineering and Technology    | 1456              | 1845:11:30        | 150595             |
| 10                                                 | Rajshahi University of Engineering and Technology | 850               | 975:38:56         | 58118              |
| 11                                                 | SUST                                              | 782               | 904:40:43         | 13224              |
| 12                                                 | Khulna University                                 | 568               | 679:13:19         | 14435              |
| 13                                                 | Bangladesh Agricultural University                | 564               | 693:15:34         | 108378             |



| SI | Institution Name                                    | Number Of Classes | Total Class Hours | Total Participants |
|----|-----------------------------------------------------|-------------------|-------------------|--------------------|
| 14 | Noakhali Science and Technology University          | 272               | 300:16:13         | 13598              |
| 15 | ICMAB                                               | 242               | 403:55:33         | 40392              |
| 16 | SAU                                                 | 241               | 238:52:13         | 16371              |
| 17 | PSTU                                                | 238               | 270:58:46         | 9630               |
| 18 | University of Chittagong                            | 187               | 458:58:40         | 9543               |
| 19 | JUST                                                | 169               | 188:41:39         | 7199               |
| 20 | Manarat International University                    | 140               | 194:40:58         | 7578               |
| 21 | University of Rajshahi                              | 140               | 163:46:14         | 8438               |
| 22 | BdREN                                               | 133               | 161:53:07         | 1475               |
| 23 | SPFMS                                               | 114               | 229:06:49         | 6301               |
| 24 | Green University of Bangladesh                      | 112               | 135:16:58         | 3617               |
| 25 | Adamjee Cantonment College                          | 111               | 99:27:24          | 3962               |
| 26 | IUT                                                 | 102               | 113:17:40         | 2783               |
| 27 | Pabna University of Science and Technology          | 94                | 101:48:14         | 6865               |
| 28 | Sher-e-Bangla Agricultural University               | 90                | 126:38:21         | 8549               |
| 29 | Bangladesh Open University                          | 89                | 106:18:57         | 3287               |
| 30 | BTCL                                                | 80                | 139:37:24         | 3306               |
| 31 | Asian University of Bangladesh                      | 76                | 127:58:22         | 1799               |
| 32 | North Western University (Khulna)                   | 73                | 63:38:12          | 1539               |
| 33 | JNU                                                 | 72                | 115:17:13         | 1948               |
| 34 | BGC Trust University Bangladesh                     | 68                | 77:39:14          | 1710               |
| 35 | Jahangirnagar University                            | 65                | 131:37:56         | 3017               |
| 36 | International Islamic University Chittagong (IIUC)  | 63                | 80:19:10          | 2960               |
| 37 | IEDCR                                               | 62                | 113:14:12         | 1677               |
| 38 | Begum Rokeya University, Rangpur                    | 58                | 73:52:13          | 5155               |
| 39 | National University                                 | 50                | 87:00:29          | 831                |
| 40 | BICM                                                | 48                | 106:23:45         | 2067               |
| 41 | Ahsanullah University of Science and Technology     | 47                | 54:56:38          | 2004               |
| 42 | Ministry of Education                               | 40                | 78:05:59          | 1725               |
| 43 | Chittagong Medical College                          | 40                | 106:06:04         | 9696               |
| 44 | ICDDRБ                                              | 39                | 52:48:41          | 2706               |
| 45 | BGMEA University of Fashion & Technology            | 38                | 118:39:31         | 1590               |
| 46 | IVTI                                                | 37                | 91:24:01          | 3107               |
| 47 | University of Liberal Arts Bangladesh (ULAB)        | 34                | 65:48:09          | 7734               |
| 48 | fDLuDCf                                             | 29                | 50:34:53          | 1653               |
| 49 | JKKNIU                                              | 29                | 47:51:09          | 945                |
| 50 | Northern University of Business & Technology Khulna | 29                | 26:27:41          | 1294               |
| 51 | Rangamati Science and Technology University (RMSTU) | 26                | 38:49:15          | 339                |
| 52 | Data Edge                                           | 26                | 25:18:54          | 214                |
| 53 | Gono Bishwabidyalay                                 | 25                | 42:10:32          | 2167               |
| 54 | Port City international Univetsity                  | 24                | 33:13:05          | 2066               |
| 55 | Dhaka Medical College                               | 23                | 22:00:33          | 751                |
| 56 | Stamford University Bangladesh                      | 22                | 41:41:26          | 339                |
| 57 | University of Barishal                              | 22                | 27:13:35          | 1022               |
| 58 | Daffodil International University                   | 21                | 24:52:35          | 1071               |
| 59 | BARI                                                | 20                | 32:42:35          | 938                |
| 60 | International Univ. of Business Agri. and Tech.     | 20                | 39:08:47          | 1027               |
| 61 | Presidency University                               | 19                | 30:27:49          | 682                |
| 62 | SOMC                                                | 19                | 22:27:24          | 2645               |
| 63 | Bangladesh Institute of Development Studies         | 19                | 31:10:14          | 257                |

| SI  | Institution Name                                               | Number Of Classes | Total Class Hours | Total Participants |
|-----|----------------------------------------------------------------|-------------------|-------------------|--------------------|
| 64  | Chittagong Independent University                              | 18                | 22:07:11          | 1513               |
| 65  | NSU                                                            | 17                | 33:29:10          | 591                |
| 66  | Faridpur Engineering College, Faridpur                         | 17                | 11:33:13          | 661                |
| 67  | Bangladesh University of Business and Tech.                    | 17                | 28:23:24          | 629                |
| 68  | ICSB                                                           | 17                | 26:02:24          | 1923               |
| 69  | BRRI                                                           | 16                | 33:52:23          | 2830               |
| 70  | UGC                                                            | 15                | 17:48:13          | 472                |
| 71  | Bangabandhu Sheikh Mujib Medical University                    | 13                | 21:37:25          | 4243               |
| 72  | BAC                                                            | 13                | 29:58:54          | 1433               |
| 73  | Mawlana Bhashani Science and Technology Univeristy (MBSTU)     | 13                | 14:38:23          | 329                |
| 74  | Bangabandhu Sheikh Mujibur Rahman Agricultural University      | 12                | 22:43:21          | 332                |
| 75  | Dhaka Dental College                                           | 12                | 22:30:38          | 1677               |
| 76  | BSMRSTU                                                        | 11                | 18:05:12          | 162                |
| 77  | RMU                                                            | 11                | 8:22:56           | 589                |
| 78  | City University                                                | 10                | 17:43:28          | 433                |
| 79  | BARD                                                           | 9                 | 16:12:37          | 34                 |
| 80  | Comilla University (CoU)                                       | 8                 | 8:59:37           | 271                |
| 81  | Spectrum Engineering Consortium Limited                        | 8                 | 7:54:34           | 27                 |
| 82  | Islamic University, Kushtia                                    | 8                 | 6:31:01           | 154                |
| 83  | East West University (EWU)                                     | 8                 | 15:02:56          | 335                |
| 84  | BRACU                                                          | 7                 | 5:46:17           | 64                 |
| 85  | Local Government Engineering Department(LGED)                  | 7                 | 6:44:17           | 259                |
| 86  | Bangladesh Institute of Management (BIM)                       | 7                 | 8:27:48           | 73                 |
| 87  | University of Asia Pacific                                     | 6                 | 7:02:16           | 252                |
| 88  | International Standard University (ISU)                        | 6                 | 10:09:40          | 207                |
| 89  | CVASU                                                          | 6                 | 5:30:37           | 223                |
| 90  | State University of Bangladesh                                 | 5                 | 7:08:15           | 1130               |
| 91  | Bangladesh Fisheries Research Institute (BFRI)                 | 5                 | 22:32:29          | 369                |
| 92  | BSEC                                                           | 5                 | 6:37:24           | 115                |
| 93  | National Institute of Textile Engineering and Research (NITER) | 4                 | 5:36:54           | 30                 |
| 94  | Independent University, Bangladesh                             | 4                 | 5:09:35           | 126                |
| 95  | Holy Family Red Crescent Medical College                       | 4                 | 4:19:39           | 575                |
| 96  | Cumilla Victoria Government College                            | 3                 | 8:50:53           | 794                |
| 97  | Eden Mohila College                                            | 3                 | 3:48:14           | 1668               |
| 98  | North East Medical College (NEMC)                              | 3                 | 5:26:04           | 14                 |
| 99  | Lalmatia Mohila College                                        | 2                 | 3:57:43           | 789                |
| 100 | Bangladesh Islami University(BIU)                              | 2                 | 2:09:41           | 25                 |
| 101 | CEDP                                                           | 2                 | 4:51:38           | 266                |
| 102 | BIGM                                                           | 1                 | 3:07:49           | 42                 |
| 103 | KWMC                                                           | 1                 | 0:48:14           | 6                  |
| 104 | Bangabandhu Digital University (BDU)                           | 1                 | 0:36:34           | 2                  |
| 105 | Kazla Technologies Limited                                     | 1                 | 0:19:44           | 7                  |
| 106 | RPSU                                                           | 1                 | 2:01:11           | 21                 |

Table 6:Institution Wise On-Prem Zoom Usage Statistics



6.4 BdREN Total Zoom Usage Statistics

|                                  |         |
|----------------------------------|---------|
| Total Number of Meetings/Classes | 115737  |
| Total Hour of Meetings/Classes   | 139956  |
| Total Number of Participants     | 5834405 |

Table 7: BdREN Total Zoom Usage Statistics

7. INSTITUTION WISE MAX HPC USAGE

| University Name | Max CPU Usage (GHz) | Max Memory Usage (GB) |
|-----------------|---------------------|-----------------------|
| DU              | 498.07              | 702.08                |
| IU              | 56.97               | 62.48                 |
| SUST            | 35.482              | 89.14                 |
| BU              | 28.96               | 15.7                  |
| BUET            | 19.7                | 52.14                 |
| JUST            | 17.87               | 79.18                 |
| JU              | 6.14                | 46.58                 |
| CVASU           | 1.332               | 20.75                 |
| BSMRAU          | 1.32                | 25.11                 |
| JKKNIU          | 0.87                | 3.82                  |
| PSTU            | 0.597               | 9.54                  |
| RUET            | 0.569               | 6.2                   |
| CUET            | 0.2                 | 5.98                  |
| PUST            | 0.197               | 13.72                 |
| MBSTU           | 0.182               | 19.07                 |
| SAU             | 0.181               | 5.67                  |
| BAU             | 0.148               | 271.15                |
| KUET            | 0.148               | 1.69                  |
| NITER           | 0.049               | 1.49                  |

Table 8: Institution Wise Max HPC Usage

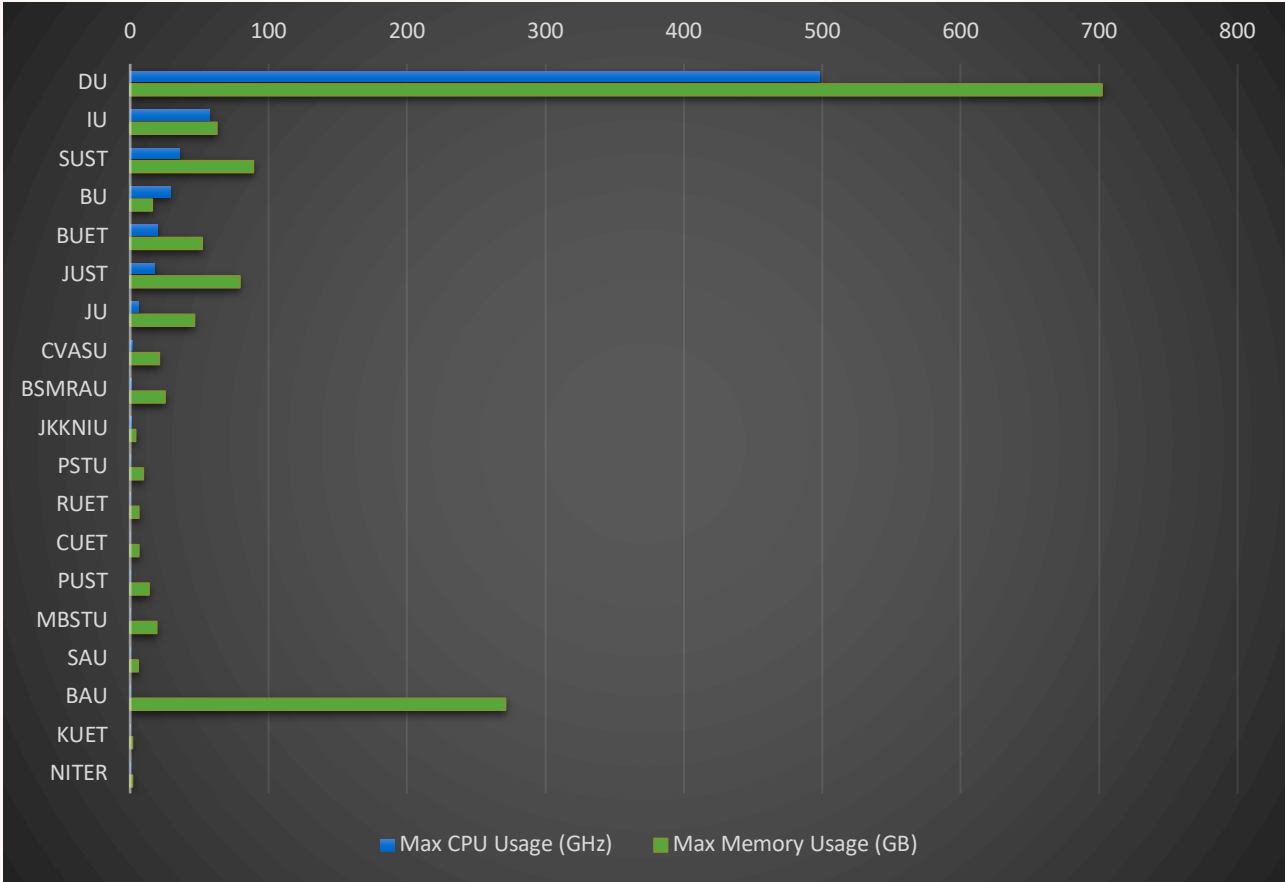


Figure 8: Institution Wise Max HPS Usage

8. ACRONYMS

|                |                                                                     |
|----------------|---------------------------------------------------------------------|
| <b>AUB</b>     | Asian University of Bangladesh                                      |
| <b>AUST</b>    | Ahsanullah University of Science & Technology                       |
| <b>AUW</b>     | Asian University for Women                                          |
| <b>BARD</b>    | Bangladesh Academy for Rural Development                            |
| <b>BARI</b>    | Bangladesh Agricultural Research Institute                          |
| <b>BAU</b>     | Bangladesh Agricultural University                                  |
| <b>BdREN</b>   | Bangladesh Research and Education Network                           |
| <b>BFRI</b>    | Bangladesh Fisheries Research Institute                             |
| <b>BIDS</b>    | The Bangladesh Institute of Development Studies                     |
| <b>BLRI</b>    | Bangladesh Livestock Research Institute                             |
| <b>BOU</b>     | Banagladesh Open University                                         |
| <b>BRACU</b>   | BRAC University                                                     |
| <b>BRRI</b>    | Bangladesh Rice Research Institute                                  |
| <b>BRUR</b>    | Begum Rokeya University, Rangpur                                    |
| <b>BSMMU</b>   | Bangabandhu Sheikh Mujib Medical University                         |
| <b>BSMRAU</b>  | Bangabandhu Sheikh Mujibur Rahman Agricultural University           |
| <b>BSMRSTU</b> | Bangabandhu Sheikh Mujibur Rahman Science and Technology University |
| <b>BTCL</b>    | Bangladesh Telecommunications Company Limited                       |
| <b>BU</b>      | University of Barishal                                              |
| <b>BUBT</b>    | Bangladesh University of Business and Technology                    |
| <b>BUET</b>    | Bangladesh University of Engineering and Technology                 |
| <b>BUP</b>     | Bangladesh University of Professionals                              |
| <b>BUTEX</b>   | Bangladesh University of Textiles                                   |
| <b>CMC</b>     | Chittagong Medical College, Chittagong                              |
| <b>CoU</b>     | Cumilla University                                                  |
| <b>CU</b>      | University of Chittagong                                            |
| <b>CUET</b>    | Chittagong University of Engineering and Technology                 |
| <b>CVASU</b>   | Chattogram Veterinary and Animal Sciences University                |
| <b>DDC</b>     | Dhaka Dental College, Dhaka                                         |
| <b>DIU</b>     | Daffodil International University                                   |
| <b>DMC</b>     | Dhaka Medical College, Dhaka                                        |
| <b>DU</b>      | University of Dhaka                                                 |
| <b>DUET</b>    | Dhaka University of Engineering and Technology                      |
| <b>EU</b>      | Eastern University                                                  |
| <b>EWU</b>     | East West University                                                |
| <b>HSTU</b>    | Hajee Mohammad Danesh Science & Technology University               |
| <b>ICDDR'b</b> | International Centre for Diarrhoeal Disease Research, Bangladesh    |
| <b>ICMAB</b>   | The Institute of Cost and Management Accountants of Bangladesh      |
| <b>ICSB</b>    | The Institute of Chartered Secretaries of Bangladesh                |
| <b>IU</b>      | Islamic University, Bangladesh                                      |
| <b>IUB</b>     | Independent University, Bangladesh                                  |
| <b>IUBAT</b>   | International University of Business Agriculture and Technology     |
| <b>IUT</b>     | Islamic University of Technology                                    |
| <b>JKKNIU</b>  | Jatiya Kabi Kazi Nazrul Islam University                            |
| <b>JnU</b>     | Jagannath University                                                |
| <b>JU</b>      | Jahangirnagar University                                            |
| <b>JUST</b>    | Jessore University of Science & Technology                          |
| <b>KU</b>      | Khulna University                                                   |

|               |                                                         |
|---------------|---------------------------------------------------------|
| <b>KUET</b>   | Khulna University of Engineering &Technology            |
| <b>LUS</b>    | Leading University,Sylhet                               |
| <b>MBSTU</b>  | Mawlana Bhashani Science and Technology University      |
| <b>MIU</b>    | Manarat International University                        |
| <b>MMC</b>    | Mymensingh Medical College, Mymensingh                  |
| <b>NINMAS</b> | National Institute of Nuclear Medicine & Allied Science |
| <b>NKN</b>    | National Knowledge Network                              |
| <b>NOC</b>    | Network Operation Center                                |
| <b>NREN</b>   | National Research and Education Network                 |
| <b>NSTU</b>   | Noakhali Science and Technology University              |
| <b>NSU</b>    | North South University                                  |
| <b>NU</b>     | National University                                     |
| <b>PSTU</b>   | Patuakhali Science and Technology University            |
| <b>PUST</b>   | Pabna University of Science and Technology              |
| <b>RAM</b>    | Random Access Memory                                    |
| <b>RMC</b>    | Rajshahi Medical College, Rajshahi                      |
| <b>RU</b>     | University of Rajshahi                                  |
| <b>RUET</b>   | Rajshahi University of Engineering & Technology         |
| <b>SAU</b>    | Sylhet Agricultural University                          |
| <b>SBAU</b>   | Sher-e-Bangla Agricultural University                   |
| <b>SBMC</b>   | Sher-e-Bangla Medical College, Barishal                 |
| <b>SOMC</b>   | Sylhet MAG Osmani Medical College, Sylhet               |
| <b>SSHMC</b>  | Shaheed Suhrawardy Medical College, Dhaka               |
| <b>SSKMC</b>  | Sheikh Sayera Khatun Medical College, Gopalganj         |
| <b>SSMC</b>   | Sir Salimullah Medical College, Dhaka                   |
| <b>SUST</b>   | Shahjalal University of Science and Technology          |
| <b>TB</b>     | Terabyte                                                |
| <b>UGC</b>    | University Grants Commission of Bangladesh              |
| <b>VU</b>     | Varendra University                                     |

